

39.0.1 UGC NET CSE | December 2012 | Part 3 | Question: 39



When a programming language has the capacity to produce new datatype, it is called as,

- A. Overloaded Language
- B. Extensible Language
- C. Encapsulated Language
- D. Abstraction Language

programming ugcnetcse-dec2012-paper3

Answer key

39.0.2 UGC NET CSE | December 2013 | Part 2 | Question: 20



The correct way to round off a floating number x to an integer value is

- A. $y = (\text{int}) (x + 0.5)$
- B. $y = \text{int} (x + 0.5)$
- C. $y = (\text{int}) x + 0.5$
- D. $y = (\text{int}) ((\text{int})x + 0.5)$

ugcnetcse-dec2013-paper2

Answer key

39.0.3 UGC NET CSE | December 2013 | Part 3 | Question: 34



_____ comparisons are necessary in the worst case to find both the maximum and minimum of n numbers.

- A. $2n - 2$
- B. $n + \text{floor}(\lg n) - 2$
- C. $\text{floor}(\frac{3n}{2}) - 2$
- D. $2 \lg n - 2$

ugcnetcse-dec2013-paper3 programming

Answer key

39.0.4 UGC NET CSE | June 2012 | Part 3 | Question: 1



Consider the following Pseudo-code segment:

$k := 0$

for $i_1 := 1$ to n

for $i_2 := 1$ to i_1

:

:

:

for $i_m := 1$ to i_{m-1}

$k := k + 1$

The value of k after the execution of this code shall be

- A. $C(n + m - 1, m)$
- B. $C(n - m + 1, m)$
- C. $C(n + m - 1, n)$
- D. $C(n - m + 1, n)$

ugcnetcse-june2012-paper3 programming algorithms

Answer key

39.0.5 UGC NET CSE | December 2012 | Part 2 | Question: 41



Enumeration is a process of

- A. Declaring a set of numbers
- B. Sorting a list of strings
- C. Assigning a legal values possible for a variable
- D. Sequencing a list of operators

ugcnetcse-dec2012-paper2 programming

Answer key

39.0.6 UGC NET CSE | June 2013 | Part 3 | Question: 50



What will be the output of the following segment of the program?

```
main()
{
char *s="hello world";
int i=7;
printf("%s", *s, i, s);
}
```

- A. syntax error B. hello w C. hello D. o world

ugcnetcse-june2013-paper3

Answer key

39.0.7 UGC NET CSE | September 2013 | Part 3 | Question: 66



The programming languages C and C++ are not strongly typed languages because

- A. Both C and C++ allow functions for which parameters are not type checked
B. Both C and C++ allow functions for which parameters are type checked
C. Both C and C++ allow functions for which parameters are not type checked and also the union types in these languages are not type checked.
D. Union types in these (C and C++) languages are not type checked

ugcnetcse-sep2013-paper3 programming

Answer key

39.0.8 UGC NET CSE | December 2013 | Part 3 | Question: 14



Which of the following concepts means adding new concepts to a program as it runs?

- A. Data hiding B. Dynamic loading
C. Dynamic typing D. Dynamic binding

ugcnetcse-dec2013-paper3 programming

Answer key

39.0.9 UGC NET CSE | December 2013 | Part 3 | Question: 15



Which of the following correctly describes overloading of functions?

- A. Virtual polymorphism B. Transient polymorphism
C. Ad-hoc polymorphism D. Pseudo polymorphism

ugcnetcse-dec2013-paper3 programming

Answer key

39.0.10 UGC NET CSE | December 2013 | Part 3 | Question: 16



Match the following with respect to the programming language :

List – I

List – II

- | | |
|---|-------------|
| a. Structured language | i. JAVA |
| b. Non-structured Language | ii. BASIC |
| c. Object oriented Programming Language | iii. PASCAL |
| d. Interpreted Programming Language | iv. FORTRAN |

Codes :

- A. a-iii, b-iv, c-i, d-ii B. a-iv, b-iii, c-ii, d-i
C. a-ii, b-iv, c-i, d-iii D. a-ii, b-iii, c-iv, d-i

ugcnetcse-dec2013-paper3 programming

Answer key 

39.0.11 UGC NET CSE | December 2015 | Part 2 | Question: 17



In C++, which system - provided function is called when no handler is provided to deal with an exception?

- A. terminate() B. unexpected() C. abort() D. kill()

ugcnetcse-dec2015-paper2 programming

Answer key 

39.0.12 UGC NET CSE | December 2015 | Part 2 | Question: 45



The ____ transfers the executable image of a C++ program from hard disk to main memory

- A. Compiler B. Linker C. Debugger D. Loader

ugcnetcse-dec2015-paper2 programming

Answer key 

39.0.13 UGC NET CSE | July 2016 | Part 2 | Question: 15



What is the value returned by the function f given below when n=100?

```
let f(int n)
{ if (n==0) then return n;
else
return n+f(n-2);
}
```

- A. 2550 B. 2556 C. 5220 D. 5520

ugcnetcse-july2016-paper2 programming

Answer key 

39.0.14 UGC NET CSE | July 2016 | Part 3 | Question: 19



Given the following statements:

- i. To implement Abstract Data Type, a programming language require a syntactic unit to encapsulate type definition
- ii. To implement ADT, a programming language requires some primitive operations that are built in the language processor
- iii. C++, Ada, Java 5.0, C#2005 provide support for parameterised ADT

Which one of the following options is correct?

- A. i, ii and iii are false B. i, ii are true, iii is false
C. i is true, ii and iii are false D. i, ii and iii are true

ugcnetcse-july2016-paper3 programming abstract-data-type

Answer key 

39.0.15 UGC NET CSE | July 2016 | Part 3 | Question: 20



Match the following types of variables with the corresponding programming languages :

- | | |
|---------------------------|-------------------------------|
| (a) Static variable | (i) Local variables in Pascal |
| (b) Stack dynamic | (ii) All variables in APL |
| (c) Explicit heap dynamic | (iii) Fortran 77 |
| (d) Implicit heap dynamic | (iv) All Objects in JAVA |

Codes :

- A. (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) B. (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
C. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) D. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Answer key **39.0.16 UGC NET CSE | July 2016 | Part 3 | Question: 21**

Which of the following is false regarding the evaluation of computer programming language:

- | | |
|----------------------------------|-------------------------------|
| A. Application oriented features | B. Efficiency and reliability |
| C. Software development | D. Hardware maintenance cost |

ugcnetcse-july2016-paper3 programming

Answer key **39.0.17 UGC NET CSE | August 2016 | Part 3 | Question: 19**

Match the following with reference to Functional programming history :

- | | |
|--|----------------------|
| a. Lambda calculus | i. Church, 1932 |
| b. Lambda calculus as programming language | ii. Wordsworth, 1970 |
| c. Lazy evaluation | iii. Haskel, 1990 |
| d. Type classes | iv. Mecarthy, 1960 |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-iv, b-i, c-iii, d-ii | B. a-i, b-iv, c-ii, d-iii |
| C. a-iii, b-ii, c-iv, d-i | D. a-ii, b-i, c-iv, d-iii |

ugcnetcse-aug2016-paper3 programming

Answer key **39.0.18 UGC NET CSE | December 2009 | Part 2 | Question: 04**

P : “Program is a step by step execution of the instructions”. Given P, which of the following is true ?

- A. Program is a subset of an instruction set.
 B. Program is a sequence of a subset of an instruction set.
 C. Program is a partially ordered set of an instruction set.
 D. All of the above

ugcnetcse-dec2009-paper2 programming

Answer key **39.0.19 UGC NET CSE | June 2009 | Part 2 | Question: 17**

The friend functions are used in situations where :

- A. We want to have access to unrelated classes
 B. Dynamic binding is required
 C. Exchange of data between classes to take place
 D. None of the above

ugcnetcse-june2009-paper2

Answer key **39.0.20 UGC NET CSE | June 2009 | Part 2 | Question: 20**

There exists a construct which returns a value ‘true’ if the argument subquery is :

- A. empty
 B. non-empty
 C. in error

D. none of the above

ugcnetcse-june2009-paper2

Answer key 

39.0.21 UGC NET CSE | December 2004 | Part 2 | Question: 14



A function object :

- A. is an instance of a class for which operator () is a member function
- B. is an instance of a class for which operator → is a member function
- C. is a pointer to any function
- D. is a member function of a class

ugcnetcse-dec2004-paper2

Answer key 

39.0.22 UGC NET CSE | December 2004 | Part 2 | Question: 12



Consider the following statements,

```
int i=4, j=3, k=0;
k=++i - --j + i++--j +j++;
```

What will be the values of i , j and k after the statement.

- A. 7,2,8
- B. 5,2,10
- C. 6,2,8
- D. 4,2,8

ugcnetcse-dec2004-paper2

Answer key 

39.0.23 UGC NET CSE | December 2006 | Part 2 | Question: 15



The declaration “unsigned u” indicates :

- A. u is an unsigned character
- B. u is an unsigned integer
- C. u is a character
- D. u is a string

ugcnetcse-dec2006-paper2

Answer key 

39.0.24 UGC NET CSE | December 2006 | Part 2 | Question: 13



```
int arr[ ]={1, 2, 3, 4}
int count;
incr( ) {return ++count;}
main( )
{
arr[count++]=incr( );
printf("arr[count]=%d\n", arr[count]);
}
```

The value printed by the above program is :

- A. 1
- B. 2
- C. 3
- D. 4

ugcnetcse-dec2006-paper2

39.0.25 UGC NET CSE | December 2007 | Part 2 | Question: 13



Which of the following is a valid C code to print character ‘A’ to ‘C’ ?

- A.
- ```
x='A' ;
switch(x)
{case 'A'= printf ("%d\n" , x);
....
case 'C'= printf ("%d\n" , x);
}
```
- B.
- ```
x='A';
switch(x)
{case 'A' <= x <= 'C' : printf ( "%d\n" , x);}
```
- C.
- ```
x='A' ;
switch(x)
{
case 'A' : printf ("%d\n" , x);
break;
case 'B' : printf ("%d\n" , x);
break;
case 'C' : printf ("%d\n" , x);
break;
}
```
- D.
- ```
x='A' ;
switch(x)
{
case 'A' =printf ( "%d\n" , x);
case 'B' =printf ( "%d\n" , x);
case 'C' =printf ( "%d\n" , x);
}
```

ugcnetcse-dec2007-paper2

Answer key 

39.0.26 UGC NET CSE | December 2007 | Part 2 | Question: 11



What cannot replace '?' in the following C-code to print all odd numbers less than 100?

```
for (i=1; ? ; i=i+2)
    printf("%d\n", i);
```

1. $i \leq 100$
2. $i \leq 101$
3. $i < 100$
4. $i < 101$

ugcnetcse-dec2007-paper2

Answer key 

39.1 Aliasing (1)

39.1.1 Aliasing: UGC NET CSE | August 2016 | Part 3 | Question: 20



Aliasing in the context of programming languages refers to :

- A. Multiple variables having the same location
- B. Multiple variables having the same identifier
- C. Multiple variables having the same value
- D. Multiple use of same variable

ugcnetcse-aug2016-paper3 programming aliasing

Answer key 

39.2 Array (3)

39.2.1 Array: UGC NET CSE | June 2013 | Part 3 | Question: 48



When an array is passed as a parameter to a function which of the following statements is correct?

- A. The function can values in the original array
- B. The function cannot values in the original array
- C. Results in compilation error
- D. Results in run-time error

ugcnetcse-june2013-paper3 programming-in-c array

Answer key

39.2.2 Array: UGC NET CSE | Junet 2015 | Part 2 | Question: 11



What is the output of the following program: (Assume that the appropriate preprocessor directives are included and there is not syntax error)

```
main( )
{
    char S[]="ABCDEFGH";
    printf("%C", *(&S[3]));
    printf("%s", S+4);
    printf("%u", S);
    /*Base address of S is 1000 */
}
```

- A. ABCDEFGH1000
- B. CDEFGH1000
- C. DDEFGHH1000
- D. DEFGH1000

ugcnetcse-june2015-paper2 programming-in-c array

Answer key

39.2.3 Array: UGC NET CSE | October 2020 | Part 2 | Question: 11



Suppose you are compiling on a machine with 1-byte chars, 2-byte shorts, 4-byte ints, and 8-byte doubles, and with alignment rules that require the address of every primitive data element to be an integer multiple of the element's size. Suppose further that the compiler is not permitted to reorder fields; padding is used to ensure alignment. How much space will be consumed by the following array?

```
struct {
    short s;
    char c;
    short t;
    char d;
    double r;
    int i;
} A[10]; /*10 element array of structs */
```

- A. 150 bytes
- B. 320 bytes
- C. 240 bytes
- D. 200 bytes

ugcnetcse-oct2020-paper2 programming-in-c array

Answer key

39.3

Binary Search Tree (1)

39.3.1 Binary Search Tree: UGC NET CSE | December 2014 | Part 3 | Question: 36



Suppose that we have numbers between 1 and 1000 in a binary search tree and we want to search for the number 365. Which of the following sequences could not be the sequence of nodes examined ?

- A. 4, 254, 403, 400, 332, 346, 399, 365
- B. 926, 222, 913, 246, 900, 260, 364, 365
- C. 927, 204, 913, 242, 914, 247, 365
- D. 4, 401, 389, 221, 268, 384, 383, 280, 365

ugcnetcse-dec2014-paper3 data-structures binary-search-tree

Answer key

39.4

Bitwise Operation (1)

39.4.1 Bitwise Operation: UGC NET CSE | December 2004 | Part 2 | Question: 11



Suppose x and y are two Integer Variables having values $0x5AB6$ and $0x61CD$ respectively. The result (in hex) of applying bitwise operator and to x and y will be :

1. $0x5089$
2. $0x4084$
3. $0x78A4$
4. $0x3AD1$

ugcnetcse-dec2004-paper2 programming-in-c bitwise-operation

Answer key

39.5 Data Types (2)

39.5.1 Data Types: UGC NET CSE | December 2005 | Part 2 | Question: 11



What is the output of the following C-program

```
main()
{
    printf("%d %d %d",size of (3.14f), size of (3.14), size of (3.141));
}
```

- A. 4 4 4 B. 4 8 10 C. 8 4 8 D. 8 8 8

ugcnetcse-dec2005-paper2 programming-in-c data-types

Answer key

39.5.2 Data Types: UGC NET CSE | June 2010 | Part 2 | Question: 15



The data type created by the data abstraction process is called

- A. Class B. Structure
C. Abstract data type D. User defined data type

ugcnetcse-june2010-paper2 programming data-types

Answer key

39.6 Dynamic Memory Allocation (1)

39.6.1 Dynamic Memory Allocation: UGC NET CSE | June 2012 | Part 3 | Question: 24



The _____ memory allocation function modifies the previous allocated space.

- A. calloc() B. free() C. malloc() D. realloc()

ugcnetcse-june2012-paper3 programming-in-c dynamic-memory-allocation

Answer key

39.7 Evaluation Postfix (1)

39.7.1 Evaluation Postfix: UGC NET CSE | December 2013 | Part 2 | Question: 21



What is the value of the postfix expression $a \ b \ c \ + \ - \ *$ (where $a=8$, $b=4$, $c=2$ and $d=5$)

- A. $-\frac{3}{8}$ B. $-\frac{8}{3}$ C. 24 D. -24

ugcnetcse-dec2013-paper2 evaluation-postfix

Answer key

39.8 Exception Handling (1)

39.8.1 Exception Handling: UGC NET CSE | December 2012 | Part 2 | Question: 29



Which of the following are two special functions that are meant for handling exception, that occur during exception handling itself?

- A. Void terminate () and void unexpected()
- B. Non void terminate () and void unexpected()
- C. Void terminate () and non void unexpected()
- D. Non void terminate () and non void unexpected()

ugcnetcse-dec2012-paper2 programming exception-handling

Answer key

39.9 Expression Evaluation (1)

39.9.1 Expression Evaluation: UGC NET CSE | December 2004 | Part 2 | Question: 13



What is the value of the arithmetic expression (Written in C)

$$2 * 3/4 - 3/4 * 2$$

- A. 0
- B. 1
- C. 1.5
- D. None of these

ugcnetcse-dec2004-paper2 programming-in-c expression-evaluation operator-precedence

Answer key

39.10 File Handling (1)

39.10.1 File Handling: UGC NET CSE | December 2012 | Part 2 | Question: 42



Which of the following mode declaration used in C++ to open a file for input?

- A. ios :: app
- B. in :: ios
- C. ios :: file
- D. ios :: in

ugcnetcse-dec2012-paper2 programming file-handling

Answer key

39.11 Graph Coloring (1)

39.11.1 Graph Coloring: UGC NET CSE | June 2012 | Part 3 | Question: 31



The upper bound of computing time of m colouring decision problem is

- A. $O(nm)$
- B. $O(n^m)$
- C. $O(nm^n)$
- D. $O(n^m m^n)$

ugcnetcse-june2012-paper3 graph-theory graph-coloring

Answer key

39.12 IO Statement (1)

39.12.1 IO Statement: UGC NET CSE | December 2008 | Part 2 | Question: 26



The output of the program code

```
main ()
{
int x=0;
while (x<=10)
for(;;)
if(++x%10==0)
break;
printf("x=%d", x);
}
```

is :

- A. x=1
- B. compilation error
- C. x=20
- D. none of the above

ugcnetcse-dec2008-paper2 programming-in-c io-statement

Answer key 

39.13 Java (4)

39.13.1 Java: UGC NET CSE | December 2012 | Part 2 | Question: 21



Which API is used to draw a circle?

- A. Circle ()
- B. Ellipse ()
- C. Round Rect ()
- D. Pie ()

ugcnetcse-dec2012-paper2 programming java bad-question

Answer key 

39.13.2 Java: UGC NET CSE | July 2018 | Part 2 | Question: 7



Which of the following statements is/are true?

P: C programming language has a weak type system with static types.

Q: Java programming language has a string type system with static types

Code:

- A. P only
- B. Q only
- C. Both P and Q
- D. Neither P nor Q

ugcnetcse-july2018-paper2 programming-in-c java

Answer key 

39.13.3 Java: UGC NET CSE | June 2012 | Part 3 | Question: 26



The mechanism that binds code and data together and keeps them secure from outside world is known as

- A. Abstraction
- B. Inheritance
- C. Encapsulation
- D. Polymorphism

ugcnetcse-june2012-paper3 programming programming-in-c java object-oriented-programming

Answer key 

39.13.4 Java: UGC NET CSE | June 2012 | Part 3 | Question: 55



Match the following with respect to java.util. * class methods :

- | | |
|---------------|--------------------------------|
| (a) Bit Set | (i) Time zone getTimezone() |
| (b) Calender | (ii) int hashCode() |
| (c) Time zone | (iii) int nextInt() |
| (d) Random | (iv) Void setID(String tzName) |

- A. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- B. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- D. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

ugcnetcse-june2012-paper3 non-gatecse object-oriented-programming java match-the-following

Answer key 

39.14 Linked List (1)

39.14.1 Linked List: UGC NET CSE | December 2013 | Part 2 | Question: 22



If the queue is implemented with a linked list, keeping track of a front pointer, which of these pointers will change during an insertion into a non-empty queue?

- A. Neither of the pointer change
- B. Only front pointer changes
- C. Only rear pointer changes
- D. Both of the pointer changes

ugcnetcse-dec2013-paper2 data-structures linked-list

Answer key

39.15

Loop (1)

39.15.1 Loop: UGC NET CSE | December 2010 | Part 2 | Question: 14



Which one of the following sentences is true ?

- A. The body of a while loop is executed at least once.
- B. The body of a do ... while loop is executed at least once.
- C. The body of a do ... while loop is executed zero or more times.
- D. A for loop can never be used in place of a while loop.

ugcnetcse-dec2010-paper2 programming loop

Answer key

39.16

Non Gatecse (4)

39.16.1 Non Gatecse: UGC NET CSE | December 2012 | Part 2 | Question: 15



What is the size of the Unicode character in Windows Operating System?

- A. 8-Bits
- B. 16-Bits
- C. 32-Bits
- D. 64-Bits

ugcnetcse-dec2012-paper2 non-gatecse programming

Answer key

39.16.2 Non Gatecse: UGC NET CSE | December 2015 | Part 3 | Question: 9



Match the following w.r.t. programming languages :

List – I

List – II

- | | |
|------------|--|
| (a) JAVA | (i) Dynamically object oriented |
| (b) Python | (ii) Statistically non-object oriented |
| (c) Prolog | (iii) Statistically object oriented |
| (d) ADA | (iv) Dynamically non-object oriented |

Codes :

- | | |
|---|---|
| A. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv) | B. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv) |
| C. (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) | D. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii) |

ugcnetcse-dec2015-paper3 programming non-gatecse

Answer key

39.16.3 Non Gatecse: UGC NET CSE | June 2012 | Part 2 | Question: 38



`printf("%c", 100);`

- A. prints 100
- B. prints ASCII equivalent of 100
- C. prints garbage
- D. None of the above

non-gatecse programming-in-c ugcnetcse-june2012-paper2

Answer key

39.16.4 Non Gatecse: UGC NET CSE | June 2014 | Part 3 | Question: 10



Code blocks allow many algorithms to be implemented with the following parameters :

- A. clarity, elegance, performance
- B. clarity, elegance, efficiency
- C. elegance, performance, execution
- D. execution, clarity, performance

ugcnetjune2014iii programming non-gatecse

Answer key

39.17

Object Oriented Programming (3)

39.17.1 Object Oriented Programming: UGC NET CSE | December 2009 | Part 2 | Question: 14



Encapsulation is

- (A) Dynamic binding
- (B) A mechanism to associate the code and data.
- (C) Data abstraction
- (D) Creating new class

ugcnetcse-dec2009-paper2 programming object-oriented-programming

Answer key

39.17.2 Object Oriented Programming: UGC NET CSE | December 2009 | Part 2 | Question: 14



A _____ is a special method used to initialize the instance variable of a class.

- (A) Member function
- (B) Destructor
- (C) Constructor
- (D) Structure

ugcnetcse-dec2009-paper2 programming object-oriented-programming

Answer key

39.17.3 Object Oriented Programming: UGC NET CSE | December 2009 | Part 2 | Question: 15



Which of the statements are true ?

- I. Function overloading is done at compile time.
- II. Protected members are accessible to the member of derived class.
- III. A derived class inherits constructors and destructors.
- IV. A friend function can be called like a normal function.
- V. Nested class is a derived class.

- (A) I, II, III
- (B) II, III, V
- (C) III, IV, V
- (D) I, II, IV

ugcnetcse-dec2009-paper2 object-oriented-programming

Answer key

39.18.1 Operator Precedence: UGC NET CSE | June 2010 | Part 2 | Question: 33



Consider the following left associative operators in decreasing order of precedence :

- subtraction (highest precedence)

* multiplication

\$ exponentiation (lowest precedence)

What is the result of the following expression ?

$3 - 2 * 4 \$ | * 2 ** 3$

A. -61

B. 64

C. 512

D. 4096

ugcnetcse-june2010-paper2 programming operator-precedence

Answer key

39.19.1 Operators: UGC NET CSE | December 2005 | Part 2 | Question: 15



What will be the value of i for the following expression:

```
int f=11, i=3 ;
i += (f>3) ? i & 2:5 ;
```

A. 2

B. 5

C. 13

D. 12

ugcnetcse-dec2005-paper2 programming-in-c operators

Answer key

39.20.1 Pointers: UGC NET CSE | December 2014 | Part 2 | Question: 12



What does the following expression means ?

$\text{char } *(*(a[N]))();$

A. A pointer to a function returning array of n pointers to function returning character pointers.

B. A function return array of N pointers to functions returning pointers to characters

C. An array of n pointers to function returning pointers to characters

D. An array of n pointers to function returning pointers to functions returning pointers to characters.

ugcnetcse-dec2014-paper2 programming-in-c pointers

Answer key

39.20.2 Pointers: UGC NET CSE | June 2010 | Part 2 | Question: 12



What will be the output of the following c-code?

```
void main ()
{
    char *p= "ayqm";
    char c;
    c=++*p;
    printf ("%c" , c);
}
```

A. a

B. c

C. b

D. q

ugcnetcse-june2010-paper2 programming-in-c pointers

Answer key 

39.21

Programming In C (15)

39.21.1 Programming In C: Modified UGC NET 2012 Memory allocation scheme



Which among the following is (are) able to modify previously allocated space

I. calloc() II. free() III. realloc()

A. Both II and III

B. I, II and III

C. III only

D. II only

programming-in-c

Answer key 

39.21.2 Programming In C: UGC NET CSE | December 2013 | Part 2 | Question: 16



What does the following declaration mean?

`int (*ptr) [10];`

A. ptr is an array of pointers of 10 integers

B. ptr is a pointer to an array of 10 integers

C. ptr is an array of 10 integers

D. none of the above

ugcnetcse-dec2013-paper2 programming-in-c

Answer key 

39.21.3 Programming In C: UGC NET CSE | December 2013 | Part 2 | Question: 17



Which of the following has compilation error in C?

A. `int n=32 ;`

B. `char ch=65 ;`

C. `float f=(float) 3.2 ;`

D. none of the above

ugcnetcse-dec2013-paper2 programming-in-c

Answer key 

39.21.4 Programming In C: UGC NET CSE | December 2014 | Part 2 | Question: 11



What will be the output of the following 'C' code ?

```
main ()
{
    int x = 128;
    printf ("\n%d", 1+x++);
}
```

A. 128

B. 129

C. 130

D. 131

ugcnetcse-dec2014-paper2 programming-in-c

Answer key 

39.21.5 Programming In C: UGC NET CSE | July 2016 | Part 2 | Question: 11



Given i=0, j=1, k=-1, x=0.5, y=0.0, what is the output of the following expression in C language?

`x*y<i+j ||k`

A. -1

B. 0

C. 1

D. 2

ugcnetcse-july2016-paper2 programming-in-c

Answer key 

39.21.6 Programming In C: UGC NET CSE | July 2016 | Part 2 | Question: 12



The following statement in C : `int (*f())[ ]`; declares

- A. a function returning a pointer to an array of integers
- B. a function returning an array of pointers to integers
- C. array of functions returning pointers to integers
- D. an illegal statement

ugcnetcse-july2016-paper2 programming-in-c

Answer key 

39.21.7 Programming In C: UGC NET CSE | June 2013 | Part 3 | Question: 33



Refer the points as listed below:

- i. What are the operator precedence rules?
- ii. What are the operator associativity rules?
- iii. What is the order of operand evaluation?
- iv. Are these restrictions on operand evaluations side effects?

Which of the above must be considered as primary design issues for arithmetic expressions?

- A. i, ii and iii
- B. i, iii and iv
- C. i, ii and iv
- D. i, ii, iii and iv

ugcnetcse-june2013-paper3 programming-in-c

Answer key 

39.21.8 Programming In C: UGC NET CSE | June 2013 | Part 3 | Question: 52



Match the following :

- | | |
|---------------------------|---|
| a. <code>calloc()</code> | i. Frees previously allocated space |
| b. <code>free()</code> | ii. Modifies previously allocated space |
| c. <code>malloc()</code> | iii. Allocates space for array |
| d. <code>realloc()</code> | iv. Allocates requested size of space |

Codes :

- A. a-iii, b-i, c-iv, d-ii
- B. a-iii, b-ii, c-i, d-iv
- C. a-iii, b-iv, c-i, d-ii
- D. a-iv, b-ii, c-iii, d-i

ugcnetcse-june2013-paper3 programming-in-c

Answer key 

39.21.9 Programming In C: UGC NET CSE | June 2014 | Part 2 | Question: 42



When we pass an array as an argument to a function, what actually gets passed ?

- A. Address of the array
- B. Values of the elements of the array
- C. Base address of the array
- D. Number of elements of the array

ugcnetcse-june2014-paper2 programming programming-in-c

Answer key 

39.21.10 Programming In C: UGC NET CSE | June 2014 | Part 2 | Question: 43



While (87) printf("computer"); The above C statement will

- A. print "computer" 87 times
- B. print "computer" 0 times
- C. print "computer" 1 times
- D. print "computer" infinite times

Answer key

39.21.11 Programming In C: UGC NET CSE | June 2014 | Part 3 | Question: 11



Match the following with respect to the jump statements :

List – I

List – II

- | | |
|-------------|--|
| a. return | i. The conditional test and increment portions |
| b. goto | ii. A value associated with it |
| c. break | iii. Requires a label for operation |
| d. continue | iv. An exit from only the innermost loop |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-ii, b-iii, c-iv, d-i | B. a-iii, b-iv, c-i, d-ii |
| C. a-iv, b-iii, c-ii, d-i | D. a-iv, b-iii, c-i, d-ii |

ugcnetjune2014iii programming programming-in-c

Answer key

39.21.12 Programming In C: UGC NET CSE | June 2019 | Part 2 | Question: 22



Consider the following C-code fragment running on a 32-bit X86 machine:

```
typedef struct {
union {
    unsigned char a;
    unsigned short b;
} U;
unsigned char c;
} S;

S B[10];
S*p=&B[4];
S*q=&B[5];
p → U.b=0x1234;
/* structure S takes 32-bits */
```

If M is the value of $q - p$ and N is the value of $((\text{int})\&(p \rightarrow c)) - ((\text{int})p)$, then (M, N) is

- | | | | |
|----------|----------|----------|----------|
| A. (1,1) | B. (3,2) | C. (1,2) | D. (4,4) |
|----------|----------|----------|----------|

ugcnetcse-june2019-paper2 programming-in-c

Answer key

39.21.13 Programming In C: UGC NET CSE | Junet 2015 | Part 2 | Question: 13



Given that $x = 7.5, j = -1.0, n = 1.0, m = 2.0$ the value of $--x + j == x > n >= m$ is:

- | | | | |
|------|------|------|------|
| A. 0 | B. 1 | C. 2 | D. 3 |
|------|------|------|------|

ugcnetcse-june2015-paper2 programming-in-c

Answer key

39.21.14 Programming In C: UGC NET CSE | September 2013 | Part 2 | Question: 35



Assume that x and y are non-zero positive integers. What does the following program segment perform?

```
while (x!=0)
{
    if (x>y)
        x=x-y;
    else y=y-x;
    printf("%d", x)
}
```

- | | |
|--------------------------------|--------------------------------|
| A. Computes LCM of two numbers | B. Computes GCD of two numbers |
|--------------------------------|--------------------------------|

C. Divides large number with small number

D. Subtracts smaller number from larger number

ugcnetsep2013ii programming-in-c

Answer key 

39.21.15 Programming In C: UGC NET CSE | September 2013 | Part 3 | Question: 54



A program contains the following declarations and initial assignments.

```
int i=8, j=5;
```

```
double x =0.005, y=-0.1;
```

```
char c='c', d='d';
```

Determine the value of the following expressions which involve the use of library functions:

```
abs(1-2*j); log(exp(x)); toupper(d)
```

A. 2; 0.005; D

B. 1; 0.005; D

C. 2; 0.005; E

D. 1; 0.005; e

ugcnetcse-sep2013-paper3 programming-in-c

Answer key 

39.22

Pseudocode (1)

39.22.1 Pseudocode: UGC NET CSE | June 2019 | Part 2 | Question: 21



Consider the following pseudo-code fragment in which an invariant for the loop is “ $m \cdot x^k = p^n$ and $k \geq 0$ ” (here, p and n are integer variable that have been initialized):

```
/* Pre-conditions :  $p \geq 1 \wedge n \geq 0$  */
```

```
/* Assume that overflow never occurs */
```

```
int  $x = p$ ; int  $k = n$ ; int  $m = 1$ ;
```

```
while ( $k < 0$ ) {
```

```
  if ( $k$  is odd) then  $m = m \cdot x$ ;
```

```
   $x = x \cdot x$ ;
```

```
   $k = \lfloor k/2 \rfloor$ ; /* floor ( $k/2$ ) */
```

```
}
```

Which of the following must be true at the end of the while loop?

A. $x = p^n$

B. $m = p^n$

C. $p = x^n$

D. $p = m^n$

ugcnetcse-june2019-paper2 pseudocode

Answer key 

39.23

Row Major Address Calculation (1)

39.23.1 Row Major Address Calculation: UGC NET CSE | December 2014 | Part 2 | Question: 24



Consider an array $A[20, 10]$, assume 4 words per memory cell and the base address of array A is 100. What is the address of $A[11, 5]$? Assume row major storage.

A. 560

B. 565

C. 570

D. 575

ugcnetcse-dec2014-paper2 array row-major-address-calculation

Answer key 

39.24

Storage Classes In C (2)

39.24.1 Storage Classes In C: UGC NET CSE | June 2009 | Part 2 | Question: 16



Which of the following does not represent a valid storage class in 'c'?

- A. automatic
C. union

- B. static
D. extern

ugcnetcse-june2009-paper2 programming-in-c storage-classes-in-c

Answer key 

39.24.2 Storage Classes In C: UGC NET CSE | June 2014 | Part 2 | Question: 41



Match the following :

List – I

- a. Automatic Storage Class
b. Register Storage Class
c. Static Storage Class
d. External Storage class

List – II

- i. Scope of the variable is global
ii. Value of the variable persists between different function calls
iii. Value stored in memory and local to the block in which the variable is defined
iv. value stored in CPU registers

Codes :

- A. a-iii; b-iv; c-i; d-ii
C. a-iv; b-iii; c-ii; d-i

- B. a-iii; b-iv; c-ii; d-i
D. a-iv; b-iii; c-i; d-ii

ugcnetcse-june2014-paper2 programming-in-c storage-classes-in-c

Answer key 

39.25 Tree Traversal (1)

39.25.1 Tree Traversal: UGC NET CSE | Junet 2015 | Part 2 | Question: 23



Level order Traversal of a rooted Tree can be done by starting from root and performing:

- A. Breadth First Search
C. Root Search

- B. Depth First Search
D. Deep Search

ugcnetcse-june2015-paper2 data-structures tree-traversal

Answer key 

39.26 Undefined Behaviour (1)

39.26.1 Undefined Behaviour: UGC NET CSE | June 2009 | Part 2 | Question: 15



What is the value of ' ψ ' after the execution of the following code statements :

$C = 10;$

$B = ++c ++ c;$

- A. 20
B. 22
C. 23
D. None

ugcnetcse-june2009-paper2 programming-in-c undefined-behaviour bad-question

Answer key 

39.27 Union (1)

39.27.1 Union: UGC NET CSE | July 2018 | Part 2 | Question: 4



What is the output of the following 'C' program? (Assuming little – endian representation of multi-byte data in which Least Significant Byte (LSB) is stored at the lowest memory address.)

```
#include<stdio.h>
#include<stdlib.h>
/* Assume short int occupies two bytes of storage */
int main ()
{
    union saving
    {
        short int one;
        char two[2];
    };
    union saving m;
    m.two [0] =5;
    m.two [1] =2;
    printf("%d, %d, %d\n", m.two[0], m.two[1], m.one);
}/* end of main */
```

- A. 5, 2, 1282 B. 5, 2, 52 C. 5, 2, 25 D. 5, 2, 517

ugcnetcse-july2018-paper2 programming-in-c union

Answer key 

39.28 Variable Scope (4)

39.28.1 Variable Scope: UGC NET CSE | December 2008 | Part 2 | Question: 25



Which one of the following describes correctly a static variable?

- A. It cannot be initialized
 B. It is initialized once at the commencement of execution and cannot be changed at run time
 C. It retains its value during the life of the program
 D. None of the above

ugcnetcse-dec2008-paper2 programming-in-c variable-scope

Answer key 

39.28.2 Variable Scope: UGC NET CSE | June 2005 | Part 2 | Question: 11



After 3 calls of the *c* function bug () below, the values of *i* and *j* will be :

```
int j = 1;
bug ()
{
    static int i = 0; int j = 0;
    i++; j++;
    return (i);
}
```

- A. $i=0, j=0$ B. $i=3, j=3$ C. $i=3, j=0$ D. $i=3, j=1$

ugcnetcse-june2005-paper2 programming-in-c variable-scope

Answer key 

39.28.3 Variable Scope: UGC NET CSE | June 2005 | Part 2 | Question: 12



Find the output of the following “C” code :

```
Main()
{ int x=20, y=35;
  x=y++ + x++;
  y=++y + ++x;
  printf ("%d, %d\n", x, y);
}
```

- A. 55,93 B. 53,97 C. 56,95 D. 57,94

ugcnetcse-june2005-paper2 programming-in-c variable-scope

Answer key

39.28.4 Variable Scope: UGC NET CSE | June 2005 | Part 2 | Question: 15



```

Main()
{ char *str="abcde";
printf("%c", *str);
printf("%c", *str++);
printf("%c", *(str++));
printf("%s", str);
}

```

The output of the above 'C' code will be :

- A. aacbcde
- B. aacccde
- C. aabcde
- D. None of these

ugcnetcse-june2005-paper2 programming-in-c variable-scope

Answer key

Answer Keys

39.0.1	B	39.0.2	A	39.0.3	C	39.0.4	A	39.0.5	C
39.0.6	X	39.0.7	C	39.0.8	B	39.0.9	C	39.0.10	A
39.0.11	A	39.0.12	D	39.0.13	A	39.0.14	D	39.0.15	C
39.0.16	D	39.0.17	Q-Q	39.0.18	Q-Q	39.0.19	C	39.0.20	Q-Q
39.0.21	A	39.0.22	Q-Q	39.0.23	Q-Q	39.0.24	Q-Q	39.0.25	Q-Q
39.0.26	Q-Q	39.1.1	Q-Q	39.2.1	A	39.2.2	D	39.2.3	C
39.3.1	C	39.4.1	B	39.5.1	Q-Q	39.5.2	Q-Q	39.6.1	D
39.7.1	D	39.8.1	A	39.9.1	Q-Q	39.10.1	D	39.11.1	C
39.12.1	Q-Q	39.13.1	B	39.13.2	Q-Q	39.13.3	C	39.13.4	A
39.14.1	C	39.15.1	Q-Q	39.16.1	B	39.16.2	B	39.16.3	B
39.16.4	B	39.17.1	Q-Q	39.17.2	Q-Q	39.17.3	Q-Q	39.18.1	Q-Q
39.19.1	Q-Q	39.20.1	X	39.20.2	Q-Q	39.21.1	Q-Q	39.21.2	B
39.21.3	D	39.21.4	B	39.21.5	C	39.21.6	A	39.21.7	D
39.21.8	A	39.21.9	C	39.21.10	D	39.21.11	A	39.21.12	C
39.21.13	A	39.21.14	B	39.21.15	A	39.22.1	B	39.23.1	A
39.24.1	Q-Q	39.24.2	B	39.25.1	A	39.26.1	Q-Q	39.27.1	Q-Q
39.28.1	Q-Q	39.28.2	Q-Q	39.28.3	Q-Q	39.28.4	Q-Q		